

Carleman Lifting for Three Benchmark Dynamical Systems

Ten-Second Para-Seq Formulations

Para-Seq

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1 Finite-section Carleman lifting

Consider an autonomous analytic system

$$\dot{x} = f(x), \quad x(0) = x_0, \quad x \in \mathbb{R}^d, \quad (1)$$

with

$$f_r(x) = \sum_{\beta \in \mathbb{N}_0^d} c_{r,\beta} x^\beta, \quad x^\beta = \prod_{s=1}^d x_s^{\beta_s}. \quad (2)$$

For every nonzero multi-index α , define $z_\alpha = x^\alpha$. Then

$$\dot{z}_\alpha = \sum_{r=1}^d \alpha_r x^{\alpha - e_r} \dot{x}_r \quad (3)$$

$$= \sum_{r=1}^d \sum_{\beta \in \mathbb{N}_0^d} \alpha_r c_{r,\beta} x^{\alpha - e_r + \beta}. \quad (4)$$

Thus $\dot{z} = Az$ with

$$A_{\alpha,\eta} = \sum_{r=1}^d \alpha_r c_{r,\eta - \alpha + e_r}, \quad (5)$$

where coefficients with negative exponents are zero. For

$$E_N = \{\alpha \in \mathbb{N}_0^d : 1 \leq |\alpha| \leq N\}, \quad (6)$$

the finite-section system is

$$\dot{z}^{(N)} = A^{(N)} z^{(N)}, \quad z_\alpha^{(N)}(0) = x_0^\alpha, \quad (7)$$

with dimension

$$n_N(d) = \sum_{k=1}^N \binom{k+d-1}{d-1} = \binom{N+d}{d} - 1. \quad (8)$$

The degree-one block reconstructs the physical state.

1.1 Segmented propagation and periodic re-lifting

A single finite section can drift or diverge over a long conservative or oscillatory trajectory. Let $\mathcal{L}_N$ denote the monomial lift and P_1 select the degree-one block. At reset times

$$0 = \tau_0 < \tau_1 < \dots < \tau_J = T, \quad (9)$$

use

$$z_j(\tau_j) = \mathcal{L}_N(\hat{x}(\tau_j)), \quad \dot{z}_j = A^{(N)} z_j, \quad \hat{x}(t) = P_1 z_j(t). \quad (10)$$

Therefore

$$\hat{x}(\tau_{j+1}) = P_1 e^{A^{(N)}(\tau_{j+1}-\tau_j)} \mathcal{L}_N(\hat{x}(\tau_j)). \quad (11)$$

Every segment is a linear lifted ODE solved by the Para-Seq dense affine associative scan. The reset uses only the predicted first block, never the reference trajectory.

2 Lotka–Volterra predator–prey model

The nonlinear system is

$$\dot{x} = \alpha x - \beta xy, \quad \dot{y} = \delta xy - \gamma y. \quad (12)$$

For degree k , define

$$z_k = [x^k, x^{k-1}y, \dots, xy^{k-1}, y^k]^T, \quad (z_k)_j = x^{k-j}y^j. \quad (13)$$

The product rule gives

$$\frac{d}{dt}(x^{k-j}y^j) = ((k-j)\alpha - j\gamma)x^{k-j}y^j - (k-j)\beta x^{k-j}y^{j+1} + j\delta x^{k-j+1}y^j. \quad (14)$$

Hence

$$\dot{z}_k = A_{k,k} z_k + A_{k,k+1} z_{k+1}, \quad (15)$$

where

$$(A_{k,k})_{j,j} = (k-j)\alpha - j\gamma, \quad (16)$$

$$(A_{k,k+1})_{j,j} = j\delta, \quad (17)$$

$$(A_{k,k+1})_{j,j+1} = -(k-j)\beta. \quad (18)$$

The infinite lift is exact because the vector field is quadratic. The notebook uses $N = 10$, dimension 65, and re-lifting every two time steps.

3 Nonlinear pendulum

With $\omega_0^2 = g/\ell$,

$$\dot{q} = v, \quad \dot{v} = -\omega_0^2 \sin q, \quad \sin q = \sum_{r=0}^{\infty} \frac{(-1)^r}{(2r+1)!} q^{2r+1}. \quad (19)$$

For $i+j = k$,

$$\frac{d}{dt}(q^i v^j) = i q^{i-1} v^{j+1} - j \omega_0^2 \sum_{r=0}^{\infty} \frac{(-1)^r}{(2r+1)!} q^{i+2r+1} v^{j-1}. \quad (20)$$

The kinematic term remains in degree k and the sine terms couple degree k to $k+2r$. In the ordering $z_k = [q^k, q^{k-1}v, \dots, v^k]^T$,

$$(A_{k,k})_{j,j+1} = k-j, \quad (21)$$

$$(A_{k,k+2r})_{j,j-1} = -j \omega_0^2 \frac{(-1)^r}{(2r+1)!}. \quad (22)$$

The finite section uses $N = 10$, dimension 65, and re-lifting every eight steps.

4 Equal-mass, equal-length double pendulum

Let $x = [q_1, q_2, v_1, v_2]^T$ and $d = q_1 - q_2$. With unit masses and lengths,

$$\dot{q}_1 = v_1, \quad \dot{q}_2 = v_2, \quad (23)$$

$$\dot{v}_1 = \frac{-3g \sin q_1 - g \sin(q_1 - 2q_2) - 2 \sin d(v_2^2 + v_1^2 \cos d)}{3 - \cos(2d)}, \quad (24)$$

$$\dot{v}_2 = \frac{2 \sin d(2v_1^2 + 2g \cos q_1 + v_2^2 \cos d)}{3 - \cos(2d)}. \quad (25)$$

The denominator equals 2 at the downward equilibrium, so the vector field is analytic near the origin.

4.1 Total-degree-seven Taylor model

Substitute $(q_1, q_2, v_1, v_2) \mapsto \varepsilon(q_1, q_2, v_1, v_2)$, expand through ε^7 , and set $\varepsilon = 1$. Then

$$\dot{v}_i = a_i^{[1]} + a_i^{[3]} + a_i^{[5]} + a_i^{[7]}, \quad i = 1, 2, \quad (26)$$

where each superscript is homogeneous total degree.

Acceleration 1

$$a_1^{[1]} = -2gq_1 + gq_2. \quad (27)$$

$$a_1^{[3]} = \frac{7gq_1^3}{3} - \frac{11gq_1^2q_2}{2} + 5gq_1q_2^2 - \frac{5gq_2^3}{3} - q_1v_1^2 - q_1v_2^2 + q_2v_1^2 + q_2v_2^2. \quad (28)$$

$$\begin{aligned} a_1^{[5]} = & -\frac{181gq_1^5}{60} + \frac{317gq_1^4q_2}{24} - \frac{143gq_1^3q_2^2}{6} + \frac{133gq_1^2q_2^3}{6} - \frac{32gq_1q_2^4}{3} + \frac{32gq_2^5}{15} + \frac{5q_1^3v_1^2}{3} \\ & + \frac{7q_1^3v_2^2}{6} - 5q_1^2q_2v_1^2 - \frac{7q_1^2q_2v_2^2}{2} + 5q_1q_2^2v_1^2 + \frac{7q_1q_2^2v_2^2}{2} - \frac{5q_2^3v_1^2}{3} - \frac{7q_2^3v_2^2}{6}. \end{aligned} \quad (29)$$

$$\begin{aligned} a_1^{[7]} = & \frac{9787gq_1^7}{2520} - \frac{17831gq_1^6q_2}{720} + \frac{1645gq_1^5q_2^2}{24} - \frac{7693gq_1^4q_2^3}{72} + \frac{913gq_1^3q_2^4}{9} - \frac{881gq_1^2q_2^5}{15} \\ & + \frac{173gq_1q_2^6}{9} - \frac{173gq_2^7}{63} - \frac{32q_1^5v_1^2}{15} - \frac{181q_1^5v_2^2}{120} + \frac{32q_1^4q_2v_1^2}{3} + \frac{181q_1^4q_2v_2^2}{24} - \frac{64q_1^3q_2^2v_1^2}{3} \\ & - \frac{181q_1^3q_2^2v_2^2}{12} + \frac{64q_1^2q_2^3v_1^2}{3} + \frac{181q_1^2q_2^3v_2^2}{12} - \frac{32q_1q_2^4v_1^2}{3} - \frac{181q_1q_2^4v_2^2}{24} + \frac{32q_2^5v_1^2}{15} + \frac{181q_2^5v_2^2}{120}. \end{aligned} \quad (30)$$

Acceleration 2

$$a_2^{[1]} = 2gq_1 - 2gq_2. \quad (31)$$

$$a_2^{[3]} = -\frac{10gq_1^3}{3} + 8gq_1^2q_2 - 7gq_1q_2^2 + \frac{7gq_2^3}{3} + 2q_1v_1^2 + q_1v_2^2 - 2q_2v_1^2 - q_2v_2^2. \quad (32)$$

$$\begin{aligned} a_2^{[5]} = & \frac{64gq_1^5}{15} - \frac{56gq_1^4q_2}{3} + \frac{101gq_1^3q_2^2}{3} - \frac{94gq_1^2q_2^3}{3} + \frac{181gq_1q_2^4}{12} - \frac{181gq_2^5}{60} - \frac{7q_1^3v_1^2}{3} \\ & - \frac{5q_1^3v_2^2}{3} + 7q_1^2q_2v_1^2 + 5q_1^2q_2v_2^2 - 7q_1q_2^2v_1^2 - 5q_1q_2^2v_2^2 + \frac{7q_2^3v_1^2}{3} + \frac{5q_2^3v_2^2}{3}. \end{aligned} \quad (33)$$

$$\begin{aligned}
a_2^{[7]} = & -\frac{346gq_1^7}{63} + \frac{1576gq_1^6q_2}{45} - \frac{1454gq_1^5q_2^2}{15} + \frac{1360gq_1^4q_2^3}{9} - \frac{5165gq_1^3q_2^4}{36} + \frac{1246gq_1^2q_2^5}{15} \\
& - \frac{9787gq_1q_2^6}{360} + \frac{9787gq_2^7}{2520} + \frac{181q_1^5v_1^2}{60} + \frac{32q_1^5v_2^2}{15} - \frac{181q_1^4q_2v_1^2}{12} - \frac{32q_1^4q_2v_2^2}{3} + \frac{181q_1^3q_2^2v_1^2}{60} \\
& + \frac{64q_1^3q_2^2v_2^2}{3} - \frac{181q_1^2q_2^3v_1^2}{6} - \frac{64q_1^2q_2^3v_2^2}{3} + \frac{181q_1q_2^4v_1^2}{12} + \frac{32q_1q_2^4v_2^2}{3} - \frac{181q_2^5v_1^2}{60} - \frac{32q_2^5v_2^2}{15}.
\end{aligned} \tag{34}$$

The omitted terms start at total degree nine:

$$f_{\text{exact}}(x) = f^{(7)}(x) + O(\|x\|^9) \quad (x \rightarrow 0). \tag{35}$$

4.2 Carleman lift of the seventh-order model

Writing $f_r^{(7)}(x) = \sum_{\beta} c_{r,\beta} x^{\beta}$ gives

$$\dot{z}_{\alpha} = \sum_{r=1}^4 \sum_{\beta} \alpha_r c_{r,\beta} z_{\alpha - e_r + \beta}. \tag{36}$$

Because $f^{(7)}$ contains degrees 1, 3, 5, 7,

$$\dot{z}_k = A_{k,k} z_k + A_{k,k+2} z_{k+2} + A_{k,k+4} z_{k+4} + A_{k,k+6} z_{k+6}. \tag{37}$$

For $N = 7$,

$$n_7(4) = \binom{11}{4} - 1 = 329. \tag{38}$$

The error decomposition is

$$\|\hat{x}_C - x_{\text{exact}}\| \leq \|\hat{x}_C - x_7\| + \|x_7 - x_{\text{exact}}\|, \tag{39}$$

where x_7 is the seventh-order polynomial trajectory.

5 Numerical configurations and validation

All simulations use a ten-second horizon, double precision, and 513 samples. The nonlinear path uses the repository's fixed-step RK4 transition and full DEER. The linear path uses zero-order-hold discretization and the dense PyTorch associative scan.

System	Order	Lifted dimension	Re-lift interval	DEER strategy
Lotka–Volterra	10	65	2 steps	32-step continuation
Pendulum	10	65	8 steps	full horizon
Double pendulum	7	329	8 steps	full horizon

The executed notebooks produced the following maximum Euclidean state errors against independent DOP853 references:

System	Nonlinear RK4 + DEER	Segmented Carleman
Lotka–Volterra	1.466989×10^{-6}	1.305403×10^{-9}
Pendulum	5.399407×10^{-6}	6.669904×10^{-8}
Double pendulum	5.139239×10^{-5}	1.039569×10^{-5}

For the double pendulum, the seventh-order polynomial trajectory had maximum error 4.233473×10^{-5} against the exact model. These are numerical validation results for the stated grids and initial conditions, not general error bounds.

References

- [1] A. Amini, C. Zheng, Q. Sun, and N. Motee, “Carleman linearization of nonlinear systems and its finite-section approximations,” *Discrete and Continuous Dynamical Systems - B*, vol. 30, no. 2, pp. 577–603, 2025.